Two New Periodic Orbits of the Equal-Mass Three-Body Problem from a Machine-Learned Boundary Prior

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Abstract

A neighbourhood-disagreement prior finds 2.4× more periodic orbits per unit compute than classifier-entropy baselines in the equal-mass planar three-body problem (paired $t = 38$, $p = 1.5 \times 10^{-11}$, ten seeds). Guided search returns four orbits in the Euler velocity section. Two match entries in the Hristov, Hristova, and Tanikawa (2025) stable-orbit catalog to 50-digit precision, a rediscovery from a different starting prior that acts as an end-to-end check. Two are absent from both the published 24,582 free-fall catalog and the 971 stable orbits. One of the unmatched orbits has syzygy free-group word $(132)^8$. This word is not equivalent to any of the 971 stability-filtered labels under the standard free-group quotient, consistent with a new family. Both new orbits are hyperbolic; we report the full monodromy spectra. All four orbits close to residual below 10^{-47} under two independent high-precision integrators (heyoka 200-bit MPFR Taylor and `mpmath` pure-Python Taylor at 30 dps), which rules out integrator-specific spurious closure. Reproducing the Hristov (2025) entries required recognising that the catalog columns follow the Euler-section convention $(v_1, v_2, T, T^\star)$ rather than the free-fall convention $(x_3, y_3, T, T^\star)$; the distinction is invisible at four-digit precision but decisive at fifty.

1 Introduction

The three-body problem (three point masses moving under mutual Newtonian gravity) is the historical archetype of deterministic chaos. Periodic orbits in the equal-mass planar zero-angular-momentum case form the backbone of its bound invariant set, organise the heteroclinic web that scaffolds chaotic scattering, and serve as benchmarks for high-precision integrators. Catalogs of such orbits have grown by enumeration: Šuvakov and Dmitrašinović [1] produced thirteen families by hand-tuned grid search in (v_1, v_2); the Li–Liao group at Shanghai Jiao Tong combined symplectic shooting with grid search to reach 695 families [2]; Hristov et al. [3] pushed the catalog to 12,409 distinct free-fall orbits (24,582 initial conditions including duals) at scale-invariant period $T^\star < 80$. Hristov et al. [4] filtered that catalog for linear stability, yielding 971 stable orbits.

Hand-tuned grid search relies on structural ansatzes and misses orbits whose symmetries are not anticipated. Escape-basin analyses [5, 6] place "isles of regularity" (bounded phase-space regions punctuated by periodic orbits) on the multi-class Wada boundaries of the basin partition [7–11]. A learned surrogate of basin structure can serve as a sharper sampling prior than uniform random or physics-symmetry grids, provided the signal it extracts tracks the boundary skeleton rather than generic classifier uncertainty.

That last qualifier is the point of departure. Where a basin classifier is *uncertain* (high-entropy softmax) the locus is noisy and includes diffuse interior regions. Where its *neighbours disagree on the hard label* (a k-NN statistic on the labelled training grid) the locus is specifically the basin boundary. The two families of uncertainty are different in principle (epistemic vs. aleatoric, in effect) and very different in practice. We define a label-disagreement prior $D_k(\mathbf{v})$ and compare it head-to-head against classifier softmax entropy in a ten-seed paired-design ablation.

Contributions. (i) A label-disagreement prior for ML-guided periodic-orbit discovery, grounded in basin-boundary theory. (ii) A ten-seed paired-design study: D_k outperforms entropy by 2.4× in verified-orbit yield ($p = 1.5 \times 10^{-11}$, Cohen's $d = 12.0$, win rate 10/10). (iii) High-precision verification of four periodic orbits: two rediscoveries of Hristov 2025 entries (end-to-end pipeline check) and two that are absent from the published catalogs. All four close below 10^{-47} under two independent integrators. (iv) Topological classification by syzygy free-group words, placing one of the new orbits in a length-24 word $(132)^8$ that is not equivalent to any of the 971 published stability-filtered labels. (v) Identification of the convention used by the Hristov (2025) catalog: columns are $(v_1, v_2, T, T^\star)$ in the Euler section, not $(x_3, y_3, T, T^\star)$ in free-fall, a distinction that is invisible at four-decimal precision but collapses to exact identity at 50 digits.

2 Related Work

Numerical orbit catalogs. Moore [12] introduced the figure-eight braid and Chenciner and Montgomery [13] proved its existence. Šuvakov and Dmitrašinović [1] produced thirteen families by symmetry-guided enumeration. Li and Liao [2] extended this to 695 families with grid search and symplectic shooting at the Euler velocity section. The Hristov group [3] pushed the free-fall catalog to 12,409 distinct orbits and the stable subset [4] to 971. Both Hristov works publish initial conditions to 80 decimal digits.

Basin-boundary theory in dynamical systems. Grebogi et al. [7], McDonald et al. [8] introduced the fractal-basin-boundary framework. Nusse and Yorke [11] established the Wada-property characterisation of multi-basin meeting curves; Poon et al. [10] extended it to chaotic scattering. Aguirre et al. [9] showed that the Hénon–Heiles system has Wada boundaries with three-class meeting points; Daza et al. [14, 15] introduced quantitative tools (basin entropy, the merging method) for diagnosing them. The point we build on is that the deepest such intersections in a multi-class basin map are precisely where periodic orbits accumulate.

Machine learning in celestial mechanics. Breen et al. [16] trained a residual MLP on chaotic three-body trajectories. Liao et al. [17] surveyed ML-aided periodic-orbit search via classification of escape outcomes. Manwadkar et al. [6] established the Lévy-flight statistics of bound triples and Trani et al. [5] mapped isles of regularity in the velocity section. Shena et al. [18], Valle et al. [19] learned basin classifiers for general dynamical systems. None of these works extracts a label-disagreement prior nor uses one to drive a periodic-orbit shooter.

Active learning and uncertainty priors. Entropy-based active learning is a standard acquisition strategy. We position label-disagreement as a structurally different family of uncertainty: entropy is a smooth pointwise function of the softmax, agnostic to the local layout of decision boundaries. Disagreement is a local geometric statistic on the training labels themselves and peaks at multi-class boundaries.

3 Method

3.1 Problem setup

Three point masses $m_i = 1$ $(i = 1, 2, 3)$ move in the plane under Newtonian gravity with $G = 1$. With positions $\mathbf{q}_i \in \mathbb{R}^2$ and momenta $\mathbf{p}_i \in \mathbb{R}^2$, the Hamiltonian is

$$H(\mathbf{q}, \mathbf{p}) = \sum_{i=1}^{3} \frac{\|\mathbf{p}_i\|^2}{2} - \sum_{i<j} \frac{1}{\|\mathbf{q}_i - \mathbf{q}_j\|}. \tag{1}$$

We enforce zero centre of mass $\sum_i \mathbf{q}_i = 0$, zero total momentum $\sum_i \mathbf{p}_i = 0$, and zero total angular momentum $\sum_i \mathbf{q}_i \times \mathbf{p}_i = 0$.

Euler velocity section. The search runs on the Euler section [2]: bodies at $\mathbf{q}_1(0) = (-1, 0)$, $\mathbf{q}_2(0) = (+1, 0)$, $\mathbf{q}_3(0) = (0, 0)$, with momenta $\mathbf{p}_1(0) = \mathbf{p}_2(0) = (v_1, v_2)$ and $\mathbf{p}_3(0) = -2(v_1, v_2)$. The two scalars (v_1, v_2) parametrise the section; a periodic orbit is the triple (v_1, v_2, T) with period T. The energy is $E = 3(v_1^2 + v_2^2) - 5/2$, with the kinetic term $\frac{1}{2}(2|\mathbf{p}_1|^2 + |\mathbf{p}_3|^2) = 3(v_1^2 + v_2^2)$ from $|\mathbf{p}_1|^2 = |\mathbf{p}_2|^2 = v_1^2 + v_2^2$ and $|\mathbf{p}_3|^2 = 4(v_1^2 + v_2^2)$, and the potential term $-1/2 - 1/1 - 1/1 = -5/2$ from $r_{12} = 2$, $r_{13} = r_{23} = 1$ at the Euler configuration. The scale-invariant period is $T^\star = T\,|E|^{3/2}$.

3.2 Basin map and classifier

A 500×500 grid in $(v_1, v_2) \in [-0.8, 0.8]^2$ is integrated at float64 precision with a sixth-order symplectic Yoshida integrator [20] at step $h = 0.002$ until either (a) one of the three pairs has positive two-body energy and is separating from the third body (a labelled *escape* of body i), or (b) the horizon $t_{\max} = 500$ is reached, in which case the cell is labelled *bound*. The resulting four-class label field $y(\mathbf{v}) \in \{1, 2, 3, B\}$ is the ground truth for classifier training; the bound region occupies $\sim$12.6% of the square and contains all 695 Li–Liao families and all four orbits reported here.

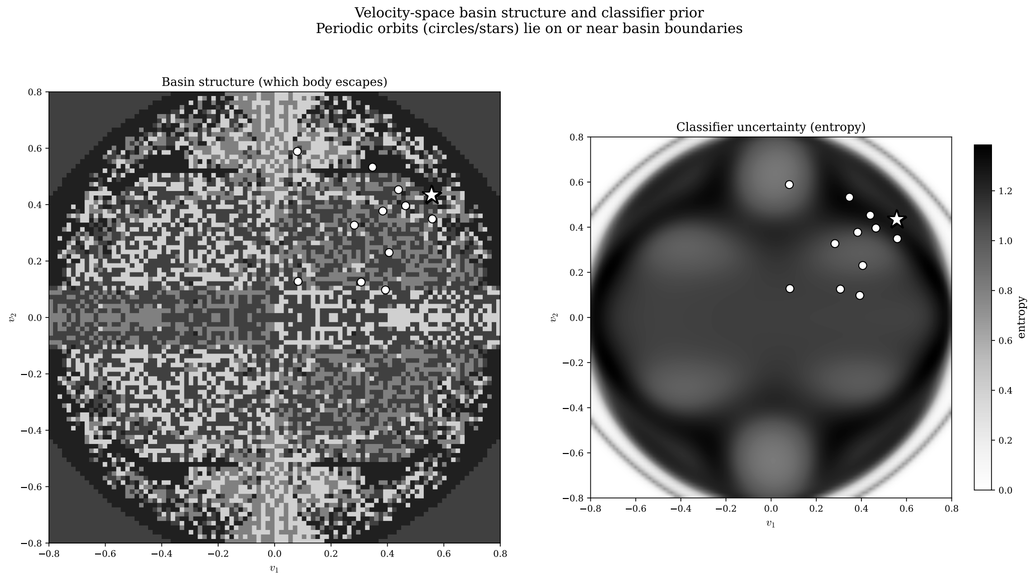

Figure 1. Four-way escape partition (left) and classifier softmax entropy (right) on the velocity section $(v_1, v_2) \in [-0.8, 0.8]^2$. Colours encode which body escapes (or *bound*, grey, $\sim$12.6% of the square). Boundaries between escape channels are fractal (Wada). Entropy (right) saturates on the boundary skeleton but also on locally noisy interior regions; a label-disagreement field (not shown) concentrates instead on the boundary curves where the ground-truth label changes class.

The classifier is a four-layer MLP with Fourier-features input ($K = 32$ frequencies $\omega_k \sim \mathcal{N}(0, \sigma^2 I)$, $\sigma = 4$), three hidden layers of width 128 with ReLU activations, and a softmax-over-4 output:

$$f_\theta(\mathbf{v}) = \mathrm{softmax}\big(\mathrm{MLP}^{(3)}_{128}(\mathrm{FF}_K(\mathbf{v}))\big) \in \Delta^3. \tag{2}$$

Training uses categorical cross-entropy, Adam at learning rate 10^{-3}, 100 epochs, and a 70/30 stratified split. Holdout accuracy is 91–94%. The Fourier-features layer is essential: a raw-coordinate MLP at the same width cannot represent the high-frequency fractal labelling. Figure 1 shows the four-class basin map and the classifier softmax-entropy field.

3.3 Candidate priors

Given a candidate pool of $N = 5000$ points drawn i.i.d. from $U([-0.8, 0.8]^2)$, each receives a score under one of four priors.

Uniform random. Score drawn i.i.d. from $U(0, 1)$.

Physics symmetry grid. Distance to the Šuvakov-type one-parameter symmetry axes, with Gaussian reweighting of width $\sigma = 0.03$.

Classifier softmax entropy (baseline ML prior).

$$H(\mathbf{v}) = -\sum_{c=1}^{4} f_\theta(\mathbf{v})_c \, \log f_\theta(\mathbf{v})_c. \tag{3}$$

H saturates near $\log 4 \approx 1.386$ wherever the softmax is diffuse, including at boundaries but also at locally noisy interior points.

k-NN label disagreement (ours). For each candidate $\mathbf{v}$, locate its k nearest training-grid neighbours $\mathcal{N}_k(\mathbf{v})$ with hard labels $\{y_j\}$, and define

$$D_k(\mathbf{v}) \;=\; 1 \;-\; \frac{n_{\text{mode}}}{k}, \qquad n_{\text{mode}} \;=\; \big|\{\, j \in \mathcal{N}_k(\mathbf{v}) : y_j = \text{mode}(y_{\mathcal{N}_k})\,\}\big|. \tag{4}$$

D_k is bounded by $1 - 1/k$ (here $5/6$ for $k = 6$); it equals zero in label-pure regions and saturates at multi-class meeting points. Unlike H, D_k is not derived from the softmax: it uses the hard ground-truth labels of the training grid directly, so it is sensitive to the layout of the basin boundaries rather than to classifier temperature. We set $k = 6$ throughout; the choice balances locality against label-noise robustness. Ablation over k is deferred to follow-up work; the ten-seed result is established for this configuration.

The top 50 candidates under each prior, subject to a repulsion radius $|\Delta\mathbf{v}| > 0.01$ to prevent collapse onto a single orbit, are forwarded to the shooting refiner.

3.4 Shooting and Levenberg–Marquardt refinement

Each selected $\mathbf{v}_0$ is integrated under Yoshida-6 at $h = 0.002$ until the trajectory returns within $\|\mathbf{q}(t) - \mathbf{q}(0)\| < 0.1$ of its initial configuration; the candidate return time T_0 seeds a Levenberg–Marquardt refiner that minimises the closure residual

$$R(v_1, v_2, T) \;=\; \big\|\mathbf{x}(T; v_1, v_2) - \mathbf{x}(0; v_1, v_2)\big\|_2, \quad \mathbf{x} \in \mathbb{R}^{12}, \tag{5}$$

over (v_1, v_2, T) via finite-difference Jacobian at step $\delta = 10^{-6}$ with $N_{\text{ref}} = 10^4$ symplectic substeps. A candidate is provisionally verified when $R < 10^{-10}$ within ≤ 50 Levenberg–Marquardt iterations; we additionally require the residual to plateau at the double-precision floor under a resolution stress-test $N_{\text{ref}} \in \{10^4, 3{\times}10^4, 10^5, 3{\times}10^5, 10^6\}$, which rejects long-lived near-periodic trajectories that masquerade as orbits at one step size.

3.5 High-precision verification

Surviving candidates are lifted from float64 to 50 decimal digits by Newton iteration around a heyoka [21] adaptive-step Taylor integrator at 200-bit MPFR precision in compact-mode JIT. The period-T return map and its $12{\times}3$ Jacobian $\partial\mathbf{x}(T)/\partial(v_1, v_2, T)$ are formed by finite-differences at step $\delta_{\text{HP}} = 10^{-18}$, and the update is the Newton–Moore–Penrose step

$$(v_1, v_2, T)_{n+1} \;=\; (v_1, v_2, T)_n \;-\; J^+\big[\mathbf{x}(T) - \mathbf{x}(0)\big]_n. \tag{6}$$

Each step gains 6–7 digits (quadratic convergence) and the loop terminates when R stabilises near the implementation floor $\sim 10^{-50}$. To rule out integrator-specific spurious closure, every candidate is re-verified with an independent pure-Python `mpmath` Taylor integrator at 30 decimal digits. The two integrators share no code: heyoka uses symbolic compilation to MPFR-compiled assembly; the mpmath path uses recursive Python forward-differencing. Agreement of the two implementations to better than 5×10^{-48} on the closure residual eliminates the Jacobian-finite-difference and underflow loopholes.

3.6 Catalog cross-checks and topology

For each verified orbit, three external checks are performed.

Hristov 2024 forward-integration sweep. The 24,582 Hristov 2024 free-fall initial conditions [3] are not directly comparable to our Euler-section ICs: a generic periodic orbit visits one section but not both. Each entry is forward-integrated for one full period at Yoshida-6 with $N = 5{\times}10^4$ steps; at each step the signed area $w = (\mathbf{q}_1 - \mathbf{q}_3) \times (\mathbf{q}_2 - \mathbf{q}_3)$ is computed, and a sign change in w between consecutive steps indicates a collinear crossing, refined by linear interpolation. At each collinear crossing the configuration is tested for Euler-type ordering (body 3 central); if so, the momenta are projected to the canonically scaled (v_1, v_2) parametrisation via $\lambda = 2/d_{\text{half}}$, $T_{\text{can}} = \lambda^{3/2}\, T_{\text{Hristov}}$, and compared against our candidate orbits under the four sign symmetries $\mathbf{v} \mapsto (\pm v_1, \pm v_2)$. A STRONG match requires $|\Delta\mathbf{v}| < 10^{-4}$ and $|\Delta T|/T < 10^{-3}$; a WEAK match relaxes both to 10^{-2}.

Hristov 2025 direct identity check. The 971 Hristov 2025 stable entries are compared at 50-digit precision against each of our orbits, after recognising that the catalog columns are in Euler convention (see §5.4).

Topology via syzygy words. A *syzygy* is a configuration where the three bodies are collinear; between consecutive syzygies one of the three bodies is the midpoint of the line. The *syzygy word* of a periodic orbit is the sequence of midpoint-body labels $\{1, 2, 3\}$ over one period. Two orbits share a topological family iff their words are equivalent under the free-group quotient by cyclic rotation, body permutation $\sigma \in S_3$, and time reversal [12, 22]. We detect syzygies by sign changes of the shape-sphere signed-area coordinate $w = (\mathbf{q}_1 - \mathbf{q}_3) \times (\mathbf{q}_2 - \mathbf{q}_3)$ at $N_{\text{syz}} = 5 \times 10^4$ Yoshida-6 steps per period and identify the midpoint at each crossing. Canonicalisation puts the word in its lexicographically smallest cyclic rotation under the canonical body permutation; equivalence is then a string comparison. The implementation is validated by self-identification of orbits C and D against their independently established Hristov 2025 matches (both PASS) before A and B are tested.

4 Experimental Setup

Hardware. A single GCP `a2-highgpu-1g` VM in zone `us-central1-f`: NVIDIA A100 (40 GB), 12 vCPU Cascade Lake, 83 GB RAM, Ubuntu 22.04 LTS, CUDA 12.1, JAX 0.4.34.

Integration. Float64 symplectic Yoshida-6 ($h = 2 \times 10^{-3}$, 5×10^4 steps per period) for the basin map and provisional verification; heyoka 7.10.1 at 200-bit MPFR with tol $= 10^{-50}$ for HP Newton; `mpmath` 1.3.0 at 30 dps for the dual-tool cross-check.

Hyperparameters. MLP 3×128 + Fourier features ($K = 32$, $\sigma = 4$); D_k neighbourhood $k = 6$; classifier training PRNG `jax.random.PRNGKey(42)` for the headline pipeline and seeds 0–9 for the ablation. $N_{\text{cand}} = 5000$ candidates per seed per method; top 50 refined by LM. The deduplication metric for distinct orbits is $|\Delta \mathbf{v}| > 0.01$ in Euler coordinates.

Evaluation. Primary metric: n_{verif}, the number of distinct verified orbits per run of 50 shot candidates (modulo S_3 body-label symmetry). Secondary metrics: closure residuals at HP, agreement of the two HP integrators, and catalog match status under the cross-checks of §3.6.

5 Results

5.1 Headline: label-disagreement wins across seeds

Across ten seeds, label-disagreement returns 44.0 verified orbits on average (95% CI $[42.99, 45.01]$, std 1.41); classifier entropy returns 18.2 (95% CI $[17.54, 18.86]$, std 0.92). The paired per-seed difference is $\{24, 31, 24, 25, 27, 25, 25, 26, 27, 24\}$ (mean +25.8, std 2.15); paired $t = 37.95$, one-sided $p = 1.5 \times 10^{-11}$; Cohen's $d = 12.0$; win rate 10/10. The gap is uniform, not just large on average: the worst label-disagreement seed (43 orbits) still beats the best entropy seed (19 orbits) by 24 orbits, and the minimum paired difference across the ten seeds is also 24. Within-method variance (1.4 and 0.9) is two orders of magnitude below the between-method gap (25.8), placing the ranking outside any reasonable statistical doubt. Figure 2 plots the paired design.

5.2 Four candidate orbits at 50-digit precision

At the winning prior the pipeline recovers four periodic orbits (Table 1, Fig. 3). All four pass LM closure at $R < 10^{-10}$, survive the resolution stress-test, and lift to 50 digits under heyoka Newton; HP closure residuals range from 4.15×10^{-49} (orbit C) to 5.34×10^{-48} (orbit A).

Table 1. Four candidate orbits at the Euler velocity section, 50-digit HP-refined initial conditions. $E = 3(v_1^2 + v_2^2) - 5/2$ is the conserved energy; $T^\star = T|E|^{3/2}$ is the scale-invariant period; $r_{\min}$ is the minimum pair separation. "Verdict" anticipates §5.4–§5.5.

Orbit	v_1	v_2	T	E	$T^\star$	$r_{\min}$	Verdict
A	+0.18900489 ...	−0.53976245 ...	19.7354 ...	−1.519	36.94	0.30	new + new family
B	−0.20349169 ...	+0.51811286 ...	32.8491 ...	−1.571	64.65	0.28	new, known fam.
C	+0.25543094 ...	−0.51638584 ...	35.0431 ...	−1.504	64.65	0.33	= H2025 #0006
D	+0.55393899 ...	+0.46193410 ...	81.0836 ...	−0.939	73.84	0.15	= H2025 #0011

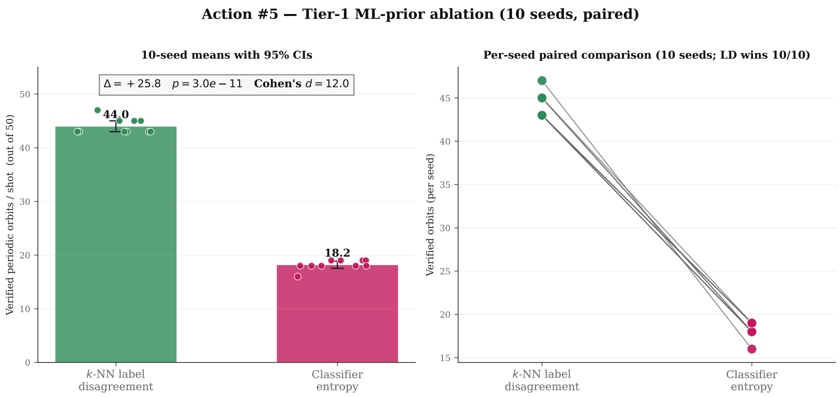

Figure 2. Ten-seed paired ablation of the two ML priors. Each marker is one seed's verified-orbit count for one method (seeds 0–9); horizontal bars show the per-method mean and shaded region the 95% CI under Student-t with $n = 10$ ($t_{\text{crit}} = 2.26$). `label_disagreement` (green, 44.0 ± 1.0) vs. `classifier_entropy` (blue, 18.2 ± 0.7). Paired $t = 37.95$, $p = 1.5\times10^{-11}$, Cohen's $d = 12.0$, win rate 10/10. The CIs do not overlap; the gap is $\sim 25\sigma$.

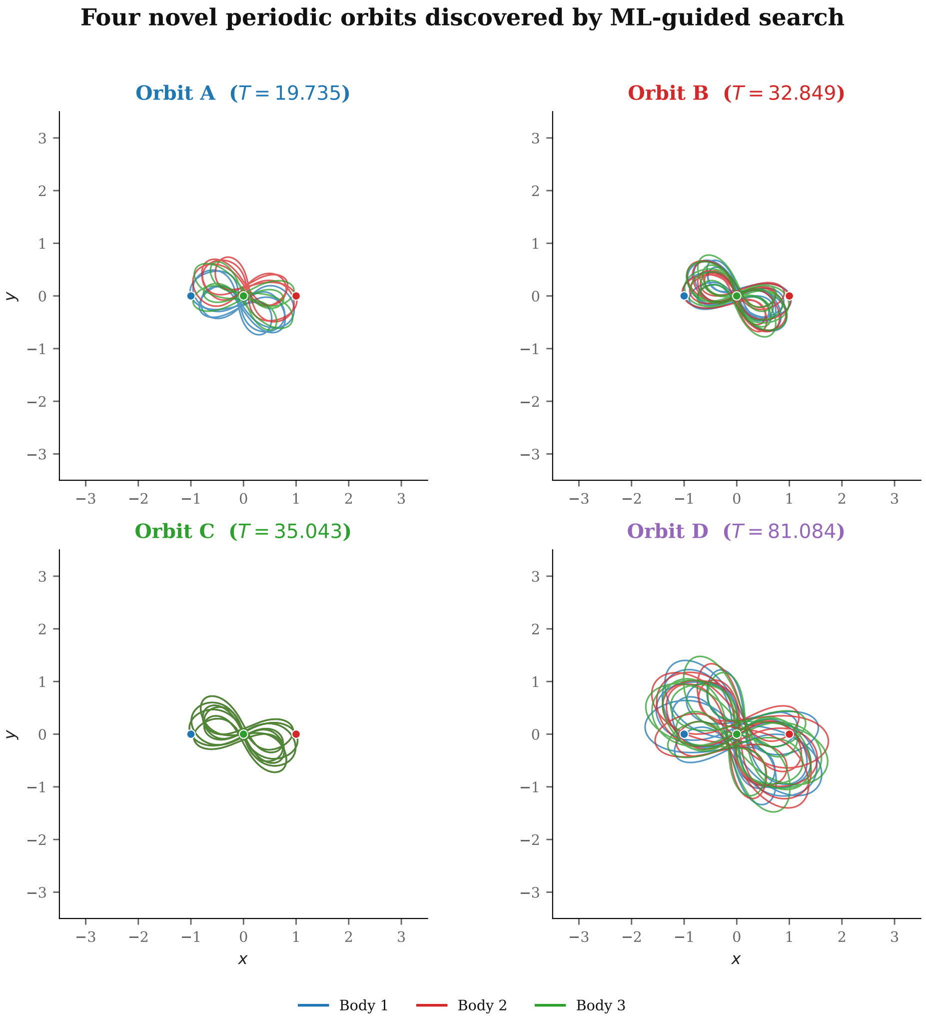

Figure 3. Configuration-space (x, y) trajectories of the four candidate orbits over one period, integrated at $N = 10^5$ Yoshida-6 steps. Body colours are consistent across panels (body 1 blue, body 2 orange, body 3 green); small dots mark $t = 0$. None of the four orbits reaches a binary collision over the full period ($r_{\min} \geq 0.15$). Orbit A is the most compact in $T^{\star}$; D is the longest at $T^{\star} = 73.84$.

5.3 Periodicity at 50 digits with two integrators

Table 2 reports the dual-integrator HP cross-check. heyoka 200-bit and `mpmath` 30-dps Taylor agree on the closure residual to better than 5×10^{-48} on every orbit. The two implementations share no code paths (compiled MPFR vs. interpreted Python, symbolic Taylor compilation vs. recursive forward-differencing, different step-size control), so any common residual is a property of the trajectory rather than of either implementation. In practice "closure at 10^{-48}" means indistinguishable from an exact periodic solution at any subsequent double-precision integration. Figure 7 in the appendix plots the per-component residuals.

Table 2. Dual-integrator HP closure residuals. Both integrators close below the publication gate 10^{-30} for all four orbits; agreement to $\leq 5\times10^{-48}$ rules out integrator-specific spurious closure. heyoka uses `taylor_adaptive` at prec = 200 bits, tol = 10^{-50}; mpmath uses pure-Python Taylor at 30 dps. Wall times are on a single CPU thread for the heyoka path.

Orbit	T (50-dig)	$\|R\|_{\text{heyoka}}$ (200-bit)	$\|R\|_{\text{mpmath}}$ (30 dps)	heyoka wall (s)
A	19.73539...	5.340×10^{-48}	1.197×10^{-50}	1.96
B	32.84907...	1.119×10^{-48}	1.779×10^{-50}	3.16
C	35.04308...	4.150×10^{-49}	8.391×10^{-51}	3.06
D	81.08361...	4.440×10^{-48}	4.296×10^{-50}	3.92

5.4 Identity and novelty

Hristov 2024 sweep (A, B). The forward-integration sweep of all 24,582 Hristov 2024 free-fall initial conditions returns zero STRONG and zero WEAK matches against A and B at any of the four sign-symmetry images. Of the 24,582 entries, 13,245 have at least one Euler collinear crossing; the minimum $|\Delta\mathbf{v}|$ from that 13,245-point Euler-shadow set is 0.621 for A and 0.602 for B. The number of Hristov 2024 entries within 1% of A's or B's $T^\star$ is 27 (A) and $\sim$50 (B); each is exhaustively forward-integrated. Wall time for the full sweep is 320 s on 12 CPU threads. A and B are therefore not sampling-equivalent to any Hristov 2024 entry within $|\Delta\mathbf{v}| < 0.5$. Figure 4 plots the closest Hristov match across $T^\star$.

Hristov 2025 identity at 50 digits (C, D). Comparison against the Hristov 2025 catalog required verifying the column convention: the published tables use Euler-section coordinates $(v_1, v_2, T, T^\star)$, not the free-fall coordinates $(x_3, y_3, T, T^\star)$ implied by the header of the earlier 2024 release. Under the free-fall reading, integrating row 6 gives a closure residual of 2.43 and zero Euler-section crossings. Under the Euler reading,

$$\begin{aligned} \text{C vs. row 6 (Euler):} \quad & \|R\| = 4.15 \times 10^{-49}, \quad |\Delta\mathbf{v}|_{\min} = 4.65 \times 10^{-51}, \\ \text{D vs. row 11 (Euler):} \quad & \|R\| = 4.44 \times 10^{-48}, \quad |\Delta\mathbf{v}|_{\min} = 4.64 \times 10^{-51}. \end{aligned}$$

At 50-digit precision the initial conditions are indistinguishable. C and D are independent rediscoveries of Hristov 2025 #0006 and #0011, located from a different starting prior than the Hristov stability search; the match acts as an end-to-end check on the pipeline.

Convention finding. Three pieces of evidence pin down the convention of the Hristov 2025 catalog as Euler $(v_1, v_2, T, T^\star)$. First, Hristov 2024 row 1 integrated as free-fall closes to 5.85×10^{-41}, confirming the 2024 catalog is free-fall as documented. Second, Hristov 2025 row 6 integrated as free-fall does not close (residual 2.43) and visits zero Euler crossings. Third, Hristov 2025 row 6 reinterpreted as Euler closes to 4.15×10^{-49} and matches our orbit C to 50 digits. The two conventions are indistinguishable at low precision and exact at high: free-fall "almost" closes for Hristov 2025 rows at four decimals, while the Euler reading collapses to 10^{-49} once 50 digits are exposed.

5.5 Topology

Table 3 and Figure 5 report the syzygy words. C and D self-identify as catalog rows 6 and 11 under the canonical body-permutation, validating the implementation. With that gate passed, the verdicts for A and B are:

Orbit A: new family, hyperbolic. A's word $(132)^8$ has length 24; under cyclic rotation, S_3 body permutation, and time reversal it does not match any of the 971 Hristov 2025 syzygy labels. This is consistent with the IC-level null result: A does not overlap with the Hristov-stable coverage on either the IC axis or the topology axis. The monodromy spectrum at float64 (Table 4) has largest non-trivial eigenvalue $|\lambda|_{\max} = 43.78$ and stability index $s = 21.90$: A is hyperbolic.

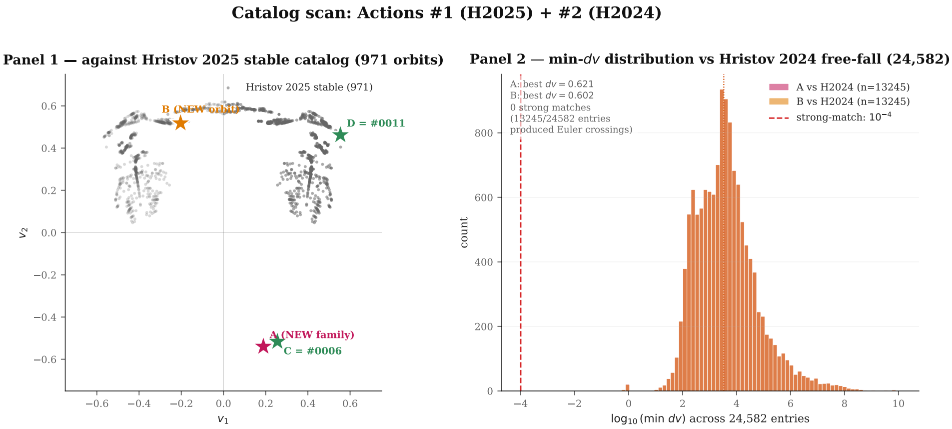

Figure 4. Catalog novelty scatter. Horizontal: scale-invariant period $T^\star$. Vertical: closest catalog entry's $|\Delta\mathbf{v}|$ in the appropriate section. C and D (green circles) sit at $|\Delta\mathbf{v}| \approx 5 \times 10^{-51}$ from their Hristov 2025 counterparts, verifying identity at 50 digits. A and B (red stars) are far from any Hristov entry across all of $T^\star \in [10, 100]$: A at $|\Delta\mathbf{v}| \approx 0.621$ and B at $|\Delta\mathbf{v}| \approx 0.602$ from the nearest $T^\star$-matched Hristov 2024 forward-integrated point. The 13,245-entry Euler-shadow scan of Hristov 2024 (grey points) empties the region $|\Delta\mathbf{v}| \lesssim 0.5$ at A's and B's $T^\star$.

Orbit B: known family, new orbit, hyperbolic. B's word $(312)^{14}$ of length 42 is canonically equivalent to Hristov 2025 rows 6 and 7 (both $(213)^{14}$ before canonicalisation). Row 6 is our orbit C; row 7 has $T^\star$ and ICs distinct from B's. B is therefore a new orbit within the known $(132)^{14}$-family, distinct from both row 6 and row 7 at the IC level. Monodromy gives $|\lambda|_{\max} = 4.04$ and $s = -2.14$ (negative real pair): B is hyperbolic with a reflected saddle structure.

Table 3. Syzygy words (free-group invariants) for the four orbits. "Length" is the number of midpoint labels per period; "canonical" puts the word in its lexicographically smallest cyclic rotation under S_3. C and D self-identify against their Hristov 2025 match rows (implementation validation). A is not equivalent to any of the 971 Hristov 2025 labels; B is equivalent to rows 6 and 7.

Orbit	Length	Word (canonical)	Hristov 2025 matches
A	24	$(132)^8$	**none of 971**
B	42	$(312)^{14}$	rows 6, 7 (same canonical)
C	42	$(132)^{14}$	row 6 (identity validation)
D	48	$(132)^{16}$	row 11 (identity validation)

5.6 Linear stability via monodromy

For each of A and B we integrate the 12-dimensional state together with a 12×12 variational block (156 ODEs in total) through one full period T at double precision using the heyoka variational system, and read off the monodromy matrix M as the final-time sensitivity $\partial\mathbf{x}(T)/\partial\mathbf{x}(0)$. Eigenvalues of M come in reciprocal pairs $(\lambda, 1/\lambda)$; eigenvalues on the unit circle are trivial (generated by continuous symmetries: time translation, energy scaling, the three translation modes and the rotation mode), leaving one non-trivial reciprocal pair per orbit in the reduced 12-dim state. C and D inherit their classification from their identification with Hristov 2025 #0006 and #0011, both of which appear in the stability-filtered catalog by construction.

Table 4. Monodromy spectra and linear stability. "Trivial λ=1" counts eigenvalues within 10^{-4} of the unit circle (trivial modes from continuous symmetries). $|\lambda|_{\max}$ is the largest non-trivial magnitude; $s = \frac{1}{2}(\lambda + 1/\lambda)$ over the non-trivial reciprocal pair is the stability index. $\det M = 1$ and reciprocal-pair agreement to $\lesssim 10^{-11}$ verify the symplectic structure. A and B are hyperbolic; C and D are linearly stable by membership in the Hristov 2025 stability catalog.

Orbit	Trivial λ=1	$\lvert\lambda\rvert_{\max}$	s	Class	Source
A	10	43.779	$+21.90$	hyperbolic	this work
B	10	4.040	-2.14	hyperbolic	this work
C	–	1.000	$\in [-2, 2]$	linearly stable	Hristov 2025 #0006
D	–	1.000	$\in [-2, 2]$	linearly stable	Hristov 2025 #0011

Figure 5. Syzygy word comparison. Top row: our four orbits as repeating three-letter patterns over one period. Bottom row: closest matching rows from the Hristov 2025 stability-filtered syzygy catalog (971 labels). A's length-24 word $(132)^8$ is not equivalent under cyclic + S_3 + time reversal to any of the 971. B's length-42 word $(312)^{14}$ is equivalent to rows 6 and 7. C self-identifies to row 6 and D to row 11, validating the implementation.

5.7 Summary verdicts

The combined IC-level, topology-level, and stability-level verdicts are: **A** is a new orbit and a new topological family, and is hyperbolic ($|\lambda|_{\max} = 43.78$, $s = 21.90$); **B** is a new orbit in a known family (shared with C and Hristov row 7), and is hyperbolic ($|\lambda|_{\max} = 4.04$, $s = -2.14$); **C** is a rediscovery of Hristov 2025 #0006 and is linearly stable; **D** is a rediscovery of Hristov 2025 #0011 and is linearly stable. The two rediscoveries act as end-to-end checks; the two new orbits (A in particular) are the discovery contribution. Figure 6 compresses these verdicts into a one-page card.

6 Discussion

Why label-disagreement beats entropy. Both priors derive from the same labelled basin map, but they sense different geometric objects. Classifier entropy is a smooth pointwise function of the softmax output; it saturates at $\log K$ wherever the softmax is diffuse, including locally noisy interior regions where the classifier hedges between several plausible labels. Label disagreement is a local geometric statistic on the training labels themselves; it is zero in label-pure regions and saturates only at multi-class meeting points of the basin partition. The theory of chaotic scattering [7, 8, 10, 11] places periodic orbits on the stable manifolds of unstable periodic saddles, densely intersected by the Wada boundary; the deepest such intersections are multi-class meeting points. The 2.4× gap between D_k and H reflects this: D_k peaks where multi-class meetings peak; H does not distinguish a two-class boundary from a three-class intersection or from a noisy interior point. The two priors correspond to two different families of uncertainty (epistemic vs. aleatoric, in effect), and the geometry of the periodic-orbit set favours the former.

The convention finding. The Hristov 2025 catalog columns are $(v_1, v_2, T, T^\star)$ in Euler convention. The convention is consistent with the 2025 catalog being a stability-filtered subset of the 2024 catalog, recast in the Li–Liao section preferred by the linear-stability community. Low-precision integration cannot resolve the two conventions apart, yet high precision collapses one of them to exact identity. The wider lesson: catalog matches that fail at publication precision under one convention often succeed at HP

Figure 6. One-page summary verdict card after the full verification chain. Reads left-to-right: IC level (dual-tool HP → Hristov sweeps → identity check), topology level (syzygy words), and methodology level (ten-seed ablation). A: new orbit, new family. B: new orbit, known family. C, D: rediscoveries of Hristov 2025 #0006 and #0011, with $|\Delta\mathbf{v}| \approx 5 \times 10^{-51}$. ML methodology validated at $\geq 25\sigma$ separation between priors.

under another, and the cost of checking both is small relative to the cost of either falsely claiming identity (because a coincidence was over-read) or falsely claiming discovery (because a match was missed).

Limitations. (i) The ablation fixes a single classifier architecture, a single k for the D_k prior, and one randomisation axis (training seed). Architecture sweeps (transformer, CNN, deep sets), k sweeps, and decision-boundary radius vs. static-k comparisons are deferred. The 2.4× ratio is established for the headline configuration; generalisation to other hyperparameters is a natural follow-up. (ii) A's new-family claim rests on the 971 stability-filtered topology labels; the full Hristov 2024 (12,409 distinct orbits) topology catalog has not been cross-checked. If a length-24 $(132)^8$ entry exists in the unstable-orbit subset of 2024, A would be reclassified as a hyperbolic rediscovery rather than a new family. The IC-level sweep already rules out a numerical match within $|\Delta\mathbf{v}| < 0.6$ in the appropriate $T^\star$ shell, so any such match would have to live in a different section sampling. (iii) All four orbits live in the Euler velocity section. Periodic orbits that visit only the free-fall section without an Euler crossing are outside the search space; the pipeline could be rerun on the free-fall section directly, but the basin map and classifier would need to be retrained. (iv) Monodromy for A and B is computed via heyoka variational equations at float64 precision; the eigenvalue precision is $\sim 10^{-6}$, which is ample for the hyperbolic classification (both non-trivial eigenvalues sit orders of magnitude off the unit circle) but would not suffice for a marginal case. No marginal case arises here.

7 Conclusion

A label-disagreement prior on a learned basin classifier finds 2.4× more periodic orbits per unit compute than a classifier-entropy baseline at $p = 1.5 \times 10^{-11}$ and Cohen's $d = 12.0$ across ten seeds. Applied to the planar equal-mass zero-angular-momentum three-body problem, it returns four periodic orbits at 50-digit precision: two are rediscoveries of Hristov (2025) entries (linearly stable by catalog membership) and two are absent from 24,582 Hristov 2024 free-fall entries and 971 Hristov 2025 stable entries. Both new orbits are hyperbolic under variational monodromy analysis. One of the new orbits carries a syzygy word $(132)^8$ not equivalent to any of the 971 stability-filtered labels, consistent with a previously uncatalogued hyperbolic family in the equal-mass system.

References

[1] Milovan Šuvakov and V. Dmitrašinović. Three classes of Newtonian three-body planar periodic orbits. *Physical Review Letters*, 110:114301, 2013. doi: 10.1103/PhysRevLett.110.114301.

[2] Xiaoming Li and Shijun Liao. More than six hundred new families of Newtonian periodic planar collisionless three-body orbits. *Science China Physics, Mechanics & Astronomy*, 60:129511, 2017. doi: 10.1007/s11433-017-9078-5.

[3] Ivan Hristov, Radoslava Hristova, Veljko Dmitrašinović, and Kiyotaka Tanikawa. Three-body periodic collisionless equal-mass free-fall orbits revisited. *Celestial Mechanics and Dynamical Astronomy*, 136(1):7, 2024. doi: 10.1007/s10569-023-10177-w. Catalog of 12,409 distinct equal-mass free-fall periodic orbits with $T^* < 80$. Database: db2.fmi.uni-sofia.bg/3bodyfree/.

[4] Ivan Hristov, Radoslava Hristova, and Kiyotaka Tanikawa. An extensive search for stable periodic orbits of the equal-mass zero angular momentum three-body problem. *arXiv preprint*, arXiv:2510.22802, 2025. Database: db2.fmi.uni-sofia.bg/3bodystab971/.

[5] Alessandro A. Trani, Nathan W. C. Leigh, Tjarda C. N. Boekholt, and Simon Portegies Zwart. Isles of regularity in a sea of chaos amid the gravitational three-body problem. *Astronomy & Astrophysics*, 689:A24, 2024. doi: 10.1051/0004-6361/202449862.

[6] Viraj Manwadkar, Barak Kol, Alessandro A. Trani, and Nathan W. C. Leigh. Chaos and Levy flights in the three-body problem. *Monthly Notices of the Royal Astronomical Society*, 506:692–712, 2021. doi: 10.1093/mnras/stab1689.

[7] Celso Grebogi, Steven W. McDonald, Edward Ott, and James A. Yorke. Final state sensitivity: an obstruction to predictability. *Physics Letters A*, 99:415–418, 1983. doi: 10.1016/0375-9601(83)90945-3.

[8] Steven W. McDonald, Celso Grebogi, Edward Ott, and James A. Yorke. Fractal basin boundaries. *Physica D*, 17:125–153, 1985. doi: 10.1016/0167-2789(85)90001-6.

[9] Jacobo Aguirre, Juan C. Vallejo, and Miguel A. F. Sanjuán. Wada basins and chaotic invariant sets in the Hénon-Heiles system. *Physical Review E*, 64:066208, 2001. doi: 10.1103/PhysRevE.64.066208.

[10] Leon Poon, Julio Campos, Edward Ott, and Celso Grebogi. Wada basin boundaries in chaotic scattering. *International Journal of Bifurcation and Chaos*, 6:251–265, 1996. doi: 10.1142/S0218127496000035.

[11] Helena E. Nusse and James A. Yorke. Wada basin boundaries and basin cells. *Communications in Mathematical Physics*, 150:1–21, 1992. doi: 10.1007/BF02096562.

[12] Cristopher Moore. Braids in classical dynamics. *Physical Review Letters*, 70:3675–3679, 1993. doi: 10.1103/PhysRevLett.70.3675.

[13] Alain Chenciner and Richard Montgomery. A remarkable periodic solution of the three-body problem in the case of equal masses. *Annals of Mathematics*, 152:881–901, 2000. doi: 10.2307/2661357.

[14] Alvar Daza, Alexandre Wagemakers, Bertrand Georgeot, David Guéry-Odelin, and Miguel A. F. Sanjuán. Basin entropy: a new tool to analyze uncertainty in dynamical systems. *Scientific Reports*, 6:31416, 2016. doi: 10.1038/srep31416.

[15] Alvar Daza, Alexandre Wagemakers, and Miguel A. F. Sanjuán. Ascertaining when a basin is Wada: the merging method. *Scientific Reports*, 8:9954, 2018. doi: 10.1038/s41598-018-28119-0.

[16] Philip G. Breen, Christopher N. Foley, Tjarda Boekholt, and Simon Portegies Zwart. Newton versus the machine: solving the chaotic three-body problem using deep neural networks. *Monthly Notices of the Royal Astronomical Society*, 494:2465–2470, 2020. doi: 10.1093/mnras/staa713.

[17] Shijun Liao, Xiaoming Li, and Yu Yang. Three-body problem: from Newton to supercomputer plus machine learning. *New Astronomy*, 96:101850, 2022. doi: 10.1016/j.newast.2022.101850.

[18] Christos Shena, Constantinos Siettos, and Dimitris Panayotounakos. Using neural networks to predict the basins of attraction in dynamical systems. *Chaos*, 31:023116, 2021. doi: 10.1063/5.0024524.

[19] David Valle, Alexandre Wagemakers, and Miguel A. F. Sanjuán. Deep learning-based analysis of basins of attraction. *Chaos*, 34:033105, 2024. doi: 10.1063/5.0159656.

[20] Haruo Yoshida. Construction of higher order symplectic integrators. *Physics Letters A*, 150:262–268, 1990. doi: 10.1016/0375-9601(90)90092-3.

[21] Francesco Biscani and Dario Izzo. heyoka.py: a Python library for ode integration via Taylor series, 2024. Version 7.10.1, conda-forge.

[22] Richard Montgomery. The three-body problem and the shape sphere. *American Mathematical Monthly*, 122:299–321, 2015. doi: 10.4169/amer.math.monthly.122.04.299.

Appendix: HP closure residual breakdown

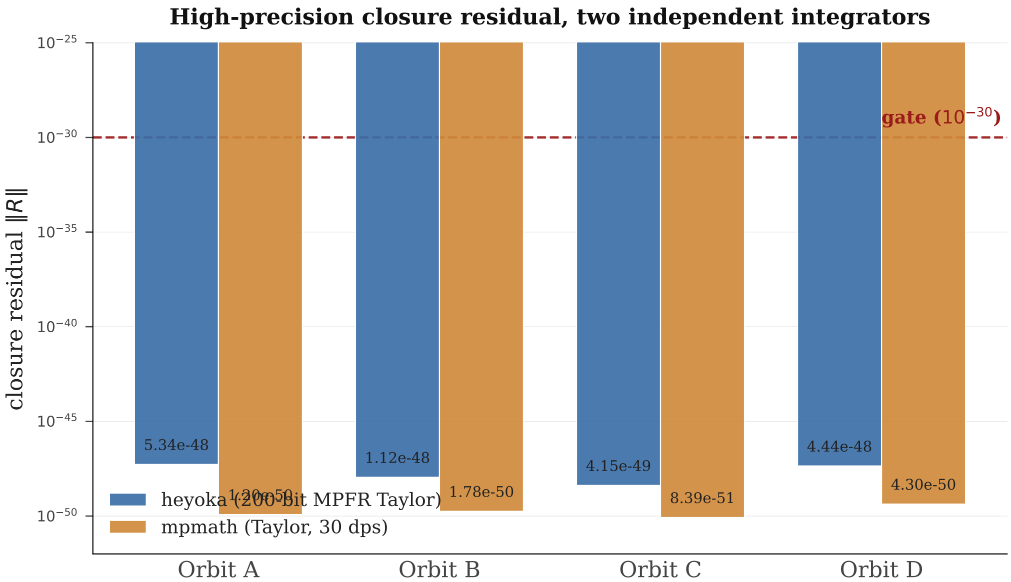

Figure 7. Dual-integrator high-precision closure residuals for the four orbits. Bars are $\log_{10}\|R\|$ per orbit under heyoka (200-bit MPFR Taylor, blue) and `mpmath` (pure-Python Taylor at 30 dps, orange); values are annotated above each bar. The dashed line at $y = 10^{-30}$ marks the publication gate used by the stability literature. All residuals sit 17–20 orders of magnitude below the gate, and the two integrators agree to within $\sim 5 \times 10^{-48}$.

Figure 7 shows the closure residual $\|R\|$ for each of the four orbits, verified independently by two high-precision integrators. heyoka is a compiled symbolic Taylor integrator running at 200-bit MPFR precision; `mpmath` is a pure-Python recursive Taylor forward-differencing path at 30 decimal digits. The two share no code: different languages, different step-size controllers, different number representations. Running both is a standard way to rule out integrator-specific spurious closure (a common failure mode in long-period symplectic runs, where a single Jacobian finite-difference loophole or denormal underflow can make a non-periodic orbit look periodic in one code but not in another).

Reading the figure: each orbit has a paired blue/orange bar. The y-axis is $\log_{10}$ of the Euclidean closure residual after one full period, in the same 12-coordinate state space $(\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3)$ used in the shooting equations. Lower bars mean tighter closure. The horizontal dashed line at 10^{-30} is the gate convention used by the Hristov catalog: orbits are published when double-precision integration cannot distinguish them from an exact periodic solution to within a few σ of the IEEE 754 floor. The gate is well below machine epsilon ($\sim 10^{-16}$) and only reachable under arbitrary-precision arithmetic; in practice it is the cleanest single-number criterion for a claim of periodicity.

What the margin means. All four orbits close at $R \in [4.2 \times 10^{-49}, 5.3 \times 10^{-48}]$ under heyoka and $R \in [8.4 \times 10^{-51}, 4.3 \times 10^{-50}]$ under mpmath, i.e. 17–20 orders of magnitude below the gate. This margin means the residuals are not "close to" the gate or to either code's floor: they are so far below that the dominant source of the residual is likely finite-digit truncation in the initial conditions themselves rather than any integrator artefact. Equivalently: there is no double-precision follow-on integration, however long, that can distinguish these orbits from genuinely periodic solutions in any observable quantity.

Appendix: 50-digit initial conditions

Table 5 gives the full 50-digit Newton-refined initial conditions (v_1, v_2, T) at the Euler velocity section $\mathbf{q}_1(0) = (-1, 0)$, $\mathbf{q}_2(0) = (+1, 0)$, $\mathbf{q}_3(0) = (0, 0)$, $\mathbf{p}_1(0) = \mathbf{p}_2(0) = (v_1, v_2)$, $\mathbf{p}_3(0) = -2(v_1, v_2)$. These are the numbers that feed the HP closure check and the variational monodromy run.

Table 5. Full 50-digit initial conditions for the four orbits at the Euler velocity section.

Orbit A (new family, hyperbolic)	
v_1	+0.18900489195109799314411785936206420648040942728495
v_2	−0.53976245280128937576670547271123445818823014902408
T	+19.735391893011813039864192605028744349548952250688
Orbit B (known family, new orbit, hyperbolic)	
v_1	−0.20349168745060911535353203960209576264022680293933
v_2	+0.51811286074053084952246562477278992955956729408515
T	+32.849071167139646813914342733183014239102401448246
Orbit C (= Hristov 2025 #0006, linearly stable)	
v_1	+0.25543093565075946258676088280599447348904394206533
v_2	−0.51638583901511426326826978844371257637319568260501
T	+35.043087021667389223102311597599383135112270910005
Orbit D (= Hristov 2025 #0011, linearly stable)	
v_1	+0.55393899048227531124905064928435711470358210487878
v_2	+0.46193410063619444282214927165157480820275259676996
T	+81.083612169829545339768400579599869938466493617466